Basics : **Encoding** : receiving sensory input and transforming it into a code or form that can be stored. **Storing** : putting coded information into memory.

Retrieval : reactivating information that has been stored in memory.

Multi Store Model of Memory - Richard Atkinson & Richard Shiffrin
Henry Molaison - Hippocampal regions

Sensory Memory ;

- Iconic memory 250 milliseconds to 1 sec
- Echoic memory 4 sec
- Decay

STM ;

- < 1 min
- 7+-2
- Recency Effect
- Chunking
- Exhaustive Scanning
- Maintenance Rehearsal
- Displacement

LTM

- No known limit
- Primacy Effect
- Faulty Organization or inappropriate retrieval or interference

Modal Model of Memory :

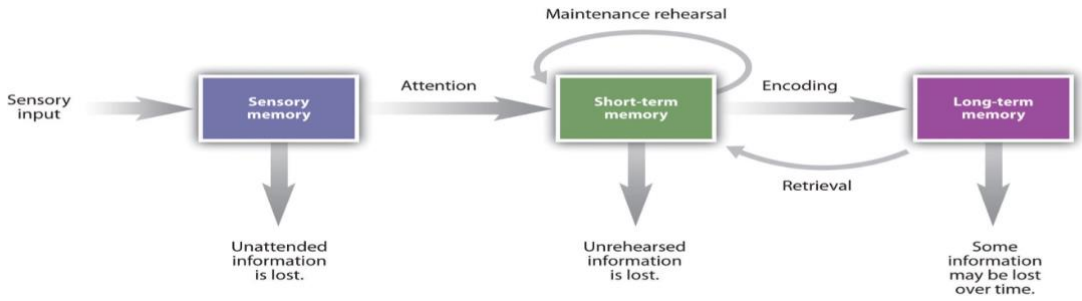
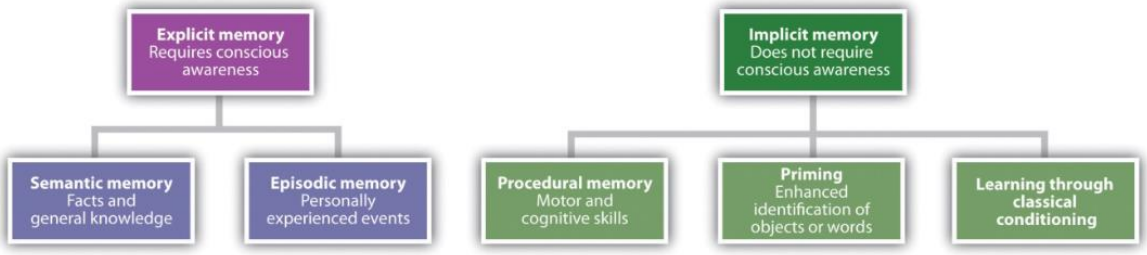
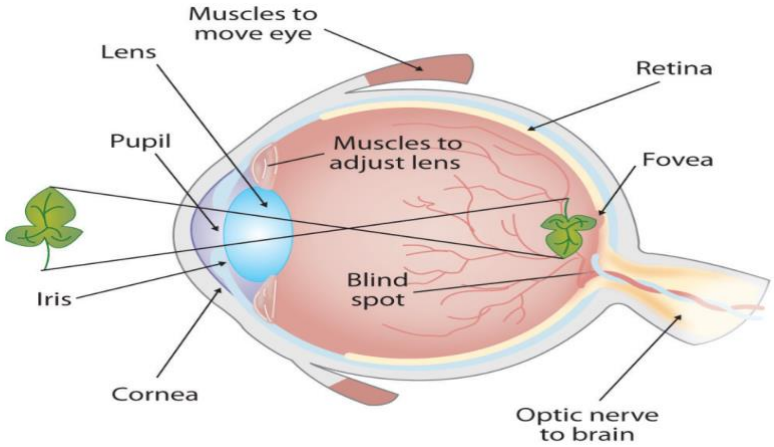


Figure 8.2 Types of Memory

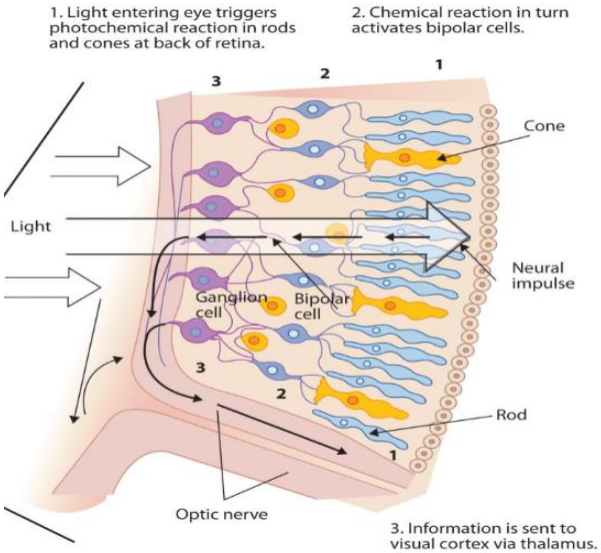


Cued Recall : retrieval of memory with the help of cues.
Flashbulb Memory : vivid, detailed memories we have of particularly momentous events.
Engram : physical encoding of information in the brain, works.
Elaboration : degree to which information is processed so that it can be tied with or integrated to existing memories.

Retroactive interference : learning something new impairs our ability to retrieve information that was learned earlier.
Proactive interference : earlier learning impairs our ability to encode information that we try to learn later.



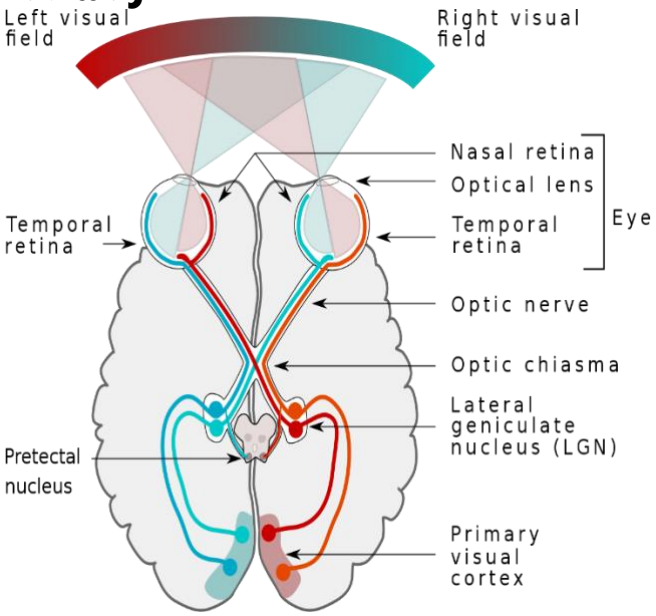
Cornea : a clear covering that protects the eye and begins to focus the incoming light.
Pupil : a small opening in the center of the eye.
Iris : the colored part of the eye that controls the size of the pupil by constricting or dilating in response to light intensity.
Lens : a structure that focuses the incoming light on the retina.
Retina : the layer of tissue at the back of the eye that contains photoreceptor cells
Fovea Centralis : a small , central pit composed of closely packed cones in the eye. It is responsible for the sharp central vision.
Optic Disc(Blind Spot) : region of the retina where the optic nerve fibers leave and the blood vessels enter and leave the retina.
Visual Accommodation: the process of changing the curvature of the lens to keep the light entering the eye focused on the retina.



Rods : cylindrical visual neurons that specialize in detecting black, white, and gray colors.
120 million rods in each eye.
Not present in fovea. Located primarily around the edges of the retina.
Peripheral vision.
scotopic vision (Dim conditions).
Contains Rhodopsin.

Cones : tapered visual neurons that are specialized in detecting fine detail and colors.
5 million cones in each eye.
Located primarily in and around the fovea.
photopic vision(bright conditions).
L, M and S cones

Visual Pathway



>90% to LGN
Remaining to Superior Colliculus and pretectum.
LGN : relay nucleus in thalamus.(not just relaying but also some processing)
Optic Radiations :
Visual cortex to cerebral cortex along what and where pathways

Color Perception

7 million colors
Hue : depend on wavelength of light.
Saturation : the degree to which a color is diluted by whiteness.
Brightness : depends on the intensity of stimulus.
Young-Helmholtz trichromatic color theory : what color we see depends on the mix of the signals from the three types of cones.
Opponent-process color theory, proposes that we analyze sensory information not in terms of three colors but rather in three sets of “opponent colors”: red-green, yellow-blue, and white-black.

Motion Perception :

- The **beta effect** refers to the perception of motion that occurs when different images are presented next to each other in succession
- The **phi phenomenon**, in which we perceive a sensation of motion caused by the appearance and disappearance of objects that are near each other.

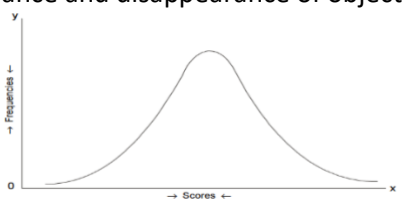


Figure 8.1 Normal curve.

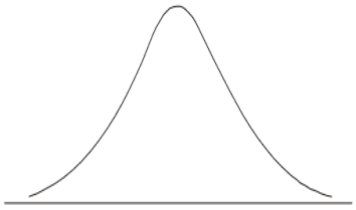


Figure 8.7 Mesokurtic.

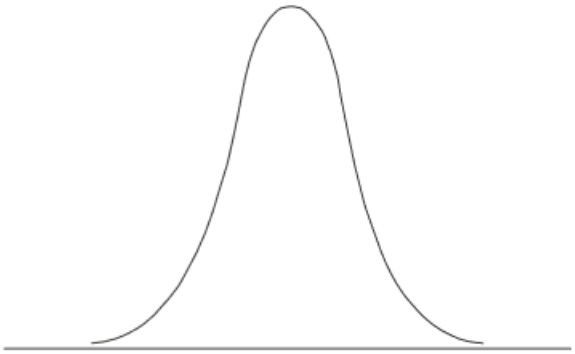


Figure 8.6 Leptokurtic.

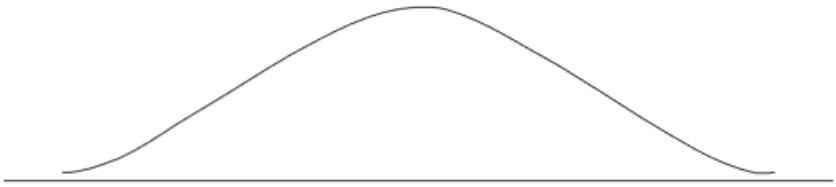


Figure 8.5 Platykurtic.

Anterior	Nearer the forward end, front of the body.
Posterior	Nearer the rear end.
Superior	Located above or higher.
Inferior	Situated below.
Proximal	Closer to the point of attachment/observation.
Distal	Remote from the point of attachment/origin/observation.
Superficial	Position closer to the surface of the body.
Deep	Position farther from the surface of the body.
Medial	Closer to the medial plane or midline.
Lateral	Away from the medial plane or midline.
Ventral	Towards the **abdomen**.
Dorsal	Towards the **back bone**.
Rostral	Towards the **beak**/nose.
Caudal	Towards the **tail**.
Cranial	Towards the **skull**.
Frontal/Coronal/Dorsal Plane	Separates the ventral and dorsal sides.
Transverse/Axial/Horizontal Plane	Separates cranial and caudal sides.
Sagittal/Anteroposterior Plane	Separates left and right.
Median Plane	Sagittal plane that is exactly in the middle of the body.
Oblique Plane	Plane that can be any angle other than horizontal or vertical.
Ipsilateral	On the same side of body/organ.
Contralateral	On the other side of body/organ.

The Brain consists of the **Cerebrum**, the **Brainstem** and the **Cerebellum**. The cerebrum, brainstem, cerebellum, and spinal cord are covered by three membranes called **Meninges**. The membranes are the tough **Dura Mater**, the middle **Arachnoid Mater** and the more delicate inner **Pia Mater**. Between the arachnoid mater and the pia mater is the **Subarachnoid Space**, which contain the **Cerebrospinal Fluid**.

The cerebrum is responsible for higher-order bodily functions such as vision, hearing, cognition, emotions, learning and voluntary movement. It is divided by the medial longitudinal fissure into two cerebral hemispheres. The hemispheres are connected by five commissures that span the longitudinal fissure, the largest of these is the corpus callosum. The outer region of the hemispheres, the **Cerebral Cortex**, is grey matter, consisting of cortical layers of neurons. The deeper subcortical regions of myelinated axons, make up the white matter.

The cerebellum sits below the cerebrum. It regulates motor functions such as fine motor control, balance, coordination and speech.

The brainstem, resembling a stalk, attaches to and leaves the cerebrum at the start of the midbrain area. The brainstem includes the midbrain, the pons, and the medulla oblongata. Behind the brainstem is the cerebellum. It controls the most lower-order bodily functions such as breathing and heart rate.

Neuroectoderm (or neural ectoderm or neural tube epithelium) - derived from ectoderm, gives rise to the nervous system.

The neuroectoderm thickens and flattens to form the **Neural Plate**. The neural plate thickens and elevates in its two edges and forms the **Neural Folds** and a **Neural Groove**. Further the neural folds converge to form the **Neural Tube**. The rest of the neural plate forms the **Neural Crest**. The Neural tube develops into four distinct regions: **Prosencephalon**, **Mesencephalon**, **Rhombencephalon** and **Spinal Cord**. Prosencephalon(or Forebrain) develops into **Telencephalon** and **Diencephalon**. Rhombencephalon(or Hindbrain) develops into **Metencephalon** and **Myelencephalon**.

The Telencephalon(or Cerebrum) gives rise to the **Cerebral Cortex**, , parts of **Basal Nuclei**, **Hippocampus** and **Olfactory Bulb**.

The Diencephalon(or interbrain) gives rise to **Thalamus**, **Hypothalamus**, **Epithalamus**, and **Subthalamus**.

The Mesencephalon(or mid brain) gives rise to **Tectum**, **Colliculi**, **Tegmentum**, **Cerebral Peduncles** and **Substantia Nigra**.

The Metencephalon gives rise to **Pons** and **Cerebellum**.

The Myelencephalon(or afterbrain) gives rise to **Medulla Oblongata**.

Major Divisions of the Brain:

Forebrain: This includes the cerebrum, thalamus, and hypothalamus.

Midbrain: It's located between the forebrain and hindbrain and plays a role in motor movement, particularly movement of the eye.

Hindbrain: It includes the cerebellum, pons, and medulla oblongata. These structures are involved in vital functions such as breathing, heart rate, and coordination.

Cortical and Subcortical Structures:

Cerebral Cortex: This is the outer layer of the cerebrum and is responsible for higher brain functions such as thought, language, and sensory perception.

Subcortical Structures:

Thalamus: Acts as a relay station for sensory information.

Hypothalamus: Regulates basic biological needs such as hunger, thirst, and temperature control, and is involved in emotional processes.

Basal Ganglia: Plays a role in motor control and procedural learning.

Limbic System: Involved in emotion, behavior, and long-term memory.

Amygdala: Part of the limbic system, responsible for processing emotions.
Hippocampus: Also part of the limbic system, involved in memory formation and spatial navigation.

Orientation and Planes of Section:

Coronal Plane: Divides the body into front (anterior) and back (posterior) sections. In brain imaging, it separates the brain into front and back portions.
Sagittal Plane: Divides the body into left and right sections. A sagittal plane that divides the brain into equal halves is called the midsagittal plane.
Transverse (Horizontal) Plane: Divides the body into upper (superior) and lower (inferior) sections. In brain imaging, it separates the brain into top and bottom portions.

Peripheral Nervous System (PNS):

Somatic Nervous System (SNS): Controls voluntary movements and transmits sensory information to the central nervous system.
Autonomic Nervous System (ANS): Regulates involuntary bodily functions, such as heart rate, digestion, and respiratory rate. It is further divided into:
Sympathetic Nervous System: Activates the body's fight-or-flight response.
Parasympathetic Nervous System: Promotes relaxation and digestion, often referred to as "rest and digest" functions.

Structure of Neurons:

- Cell Body (Soma):Contains the nucleus and other organelles necessary for cellular functions.
- Dendrites:Branch-like extensions that receive signals from other neurons or sensory receptors.
- Axon:A long, slender projection that conducts electrical impulses away from the cell body toward other neurons, muscles, or glands.
- Axon Terminal:The end of the axon where neurotransmitters are released to communicate with other neurons.

Types of Glial Cells:

- Astrocytes:Provide structural support to neurons and help regulate the extracellular environment by maintaining ion balance and neurotransmitter levels.
- Oligodendrocytes (in the central nervous system) / Schwann Cells (in the peripheral nervous system):Produce myelin, a fatty substance that insulates axons and enhances the speed of electrical signal conduction.
- Microglia:Act as the immune cells of the central nervous system, defending against pathogens and removing cellular debris.
- Ependymal Cells:Line the fluid-filled cavities (ventricles) of the brain and spinal cord, producing cerebrospinal fluid and helping to circulate it.

Action Potential:

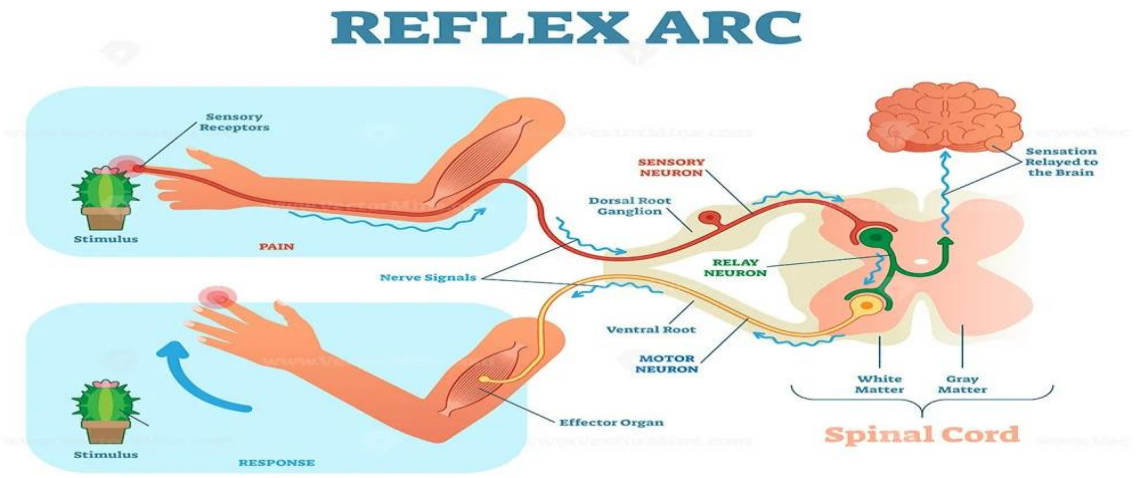
- Resting Membrane Potential:Neurons maintain an electrical gradient across their cell membranes, with a slightly negative charge inside relative to the outside (-70 millivolts in typical neurons).
- Depolarization:Stimulus causes sodium (Na+) channels to open, allowing Na+ ions to rush into the cell, making the inside less negative.
- Threshold:If the depolarization reaches a critical threshold (around -55 mV), it triggers an action potential.
- Action Potential:A rapid, transient change in membrane potential where the inside of the cell becomes briefly positive relative to the outside.
- Repolarization:After reaching its peak, voltage-gated potassium (K+) channels open, allowing K+ ions to leave the cell, restoring the negative charge.
- Hyperpolarization:The membrane potential briefly becomes more negative than the resting potential before returning to baseline.
- **Synapses:**

- Presynaptic Neuron:Neuron sending the signal. When an action potential reaches the axon terminal, it triggers the release of neurotransmitters into the synaptic cleft.
- Synaptic Cleft:The narrow gap between the presynaptic and postsynaptic neurons.
- Postsynaptic Neuron:Neuron receiving the signal. Neurotransmitters released into the synaptic cleft bind to receptors on the postsynaptic membrane, triggering changes in its electrical activity.
- **Synaptic Transmission:**
- Neurotransmitter Release:Action potential depolarizes the axon terminal, opening voltage-gated calcium (Ca2+) channels, allowing calcium ions to enter the cell.Calcium triggers the fusion of synaptic vesicles containing neurotransmitters with the presynaptic membrane, releasing neurotransmitters into the synaptic cleft via exocytosis.
- Neurotransmitter Binding and Receptor Activation:Neurotransmitters diffuse across the synaptic cleft and bind to specific receptors on the postsynaptic membrane.Binding of neurotransmitters causes ligand-gated ion channels on the postsynaptic membrane to open, leading to depolarization or hyperpolarization of the postsynaptic neuron.

Postsynaptic Response: Depending on the type of neurotransmitter and receptor, the postsynaptic neuron may become more likely (excitatory) or less likely (inhibitory) to generate an action potential.

Termination of Signal:Neurotransmitters are either degraded by enzymes, reabsorbed by the presynaptic neuron (reuptake), or diffuse away from the synapse.This process helps regulate synaptic transmission and prevent continuous signaling.

REFLEX ARC :



The receptor is present in the receptor organ.
2.The sensory neuron conducts the nerve impulses towards the central nervous system (CNS). The CNS is comprised of the brain and the spinal cord.
3. The information is transferred from one neuron to other.
4. A motor neuron conducts the response nerve from the CNS to the effector organ.
5. The effector organ shows response by contracting or secreting a product.
The instant movement performed by the individual in response to the stimulus is known as a reflex action.
Functions of Different Regions of the Brain:
Frontal Lobe:

- Motor Cortex: Controls voluntary movements and muscle coordination.
- Prefrontal Cortex: Involved in decision-making, planning, and personality expression.
- Broca's Area: Responsible for speech production and language processing.

Parietal Lobe:

- Somatosensory Cortex: Receives and processes sensory information from the body, including touch, temperature, and pain.
- Association Areas: Integrate sensory information and aid in spatial perception and navigation.

Temporal Lobe:

- Auditory Cortex: Processes auditory information, including sound recognition and language comprehension.
- Wernicke's Area: Involved in language comprehension and understanding speech.

Occipital Lobe:

Visual Cortex: Processes visual information received from the eyes, enabling sight and visual perception.

Cerebellum:

- Coordinates voluntary movements, balance, and posture.
- Plays a role in motor learning and fine motor control.

Brainstem (including the midbrain, pons, and medulla oblongata):

- Regulates vital functions such as heart rate, breathing, and blood pressure.
- Contains nuclei involved in sensory and motor processing, as well as cranial nerve functions.

Thalamus:

- Acts as a relay station for sensory information, directing signals to the appropriate areas of the cerebral cortex for processing.
- Plays a role in regulating consciousness and alertness.

Hypothalamus:

Regulates basic biological functions such as hunger, thirst, body temperature, and sleep-wake cycles.

- Controls the release of hormones from the pituitary gland, coordinating the body's response to stress and maintaining homeostasis.

Reflex Arc:

The reflex arc is a neural pathway that controls reflex actions. It involves the following components:

Receptor:

Specialized sensory neurons or cells that detect stimuli from the environment or within the body.

Sensory Neuron:

Transmits signals from the receptor to the central nervous system (spinal cord or brainstem).

Integration Center:

In the spinal cord or brainstem, processes sensory input and generates an appropriate motor response.

Motor Neuron:

Transmits signals from the integration center to an effector (muscle or gland) to produce a response.

Effector: Carries out the response, such as muscle contraction (in the case of a motor response) or hormone secretion (in the case of a gland).

Receptor Types:

Mechanoreceptors:

Detect mechanical stimuli such as pressure, touch, vibration, and stretch. Examples include receptors in the skin and hair cells in the inner ear.

Chemoreceptors:

Respond to chemical stimuli such as taste, smell, and changes in blood chemistry (e.g., oxygen and carbon dioxide levels). Examples include taste buds and olfactory receptors.

Photoreceptors:
Detect light stimuli and are found in the retina of the eye. They allow for vision and the perception of light and color.
Thermoreceptors:
Detect changes in temperature, allowing the body to sense and respond to variations in environmental temperature and regulate internal temperature.
Nociceptors:
Specialized receptors that respond to pain and tissue damage, signaling potential harm to the body.
:

Absolute Threshold:

Definition: The absolute threshold refers to the minimum amount of stimulus energy needed to detect a stimulus 50% of the time.
Example: In vision, the absolute threshold might be the minimum amount of light required for a person to see a faint star in the night sky.
Formula: There isn't a specific formula for absolute threshold, but it's often determined experimentally
through psychophysical methods.

Difference Threshold (or Just Noticeable Difference – JND):

Definition: The difference threshold is the smallest change in stimulation that can be detected 50% of the time.
Example: If you're holding two weights in your hand and can just barely tell the difference between them, that's the difference threshold.
Formula: $\Delta I = kI$, where ΔI is the difference threshold, I is the initial stimulus intensity, and k is a constant typically referred to as Weber's fraction.

Weber's Law:
Definition: Weber's law states that the just noticeable difference (JND) between two stimuli is proportional to the magnitude of the stimuli.
Example: If you're lifting weights and you can barely tell the difference between a 10-pound weight and a 11-pound weight, but can easily tell the difference between a 50-pound weight and a 55-pound weight, that follows Weber's law.
Formula: $\Delta I/I = k$, where ΔI is the difference threshold, I is the initial stimulus intensity, and k is the Weber fraction (a constant typically between 0 and 1).

Fechner's Law:
Definition: Fechner's law describes the relationship between the intensity of a stimulus and the perception of that stimulus. It states that the magnitude of subjective sensation increases proportionally to the logarithm of the stimulus intensity.
Example: If you're increasing the brightness of a light source, initially doubling the intensity might result in a noticeable increase in brightness. However, as the intensity continues to double, the perceived increase in brightness becomes smaller and smaller.

Associative Learning:

Definition: Associative learning is a type of learning principle that states that ideas and experiences reinforce each other and can be linked to one another.

Examples: Classical conditioning, where a neutral stimulus becomes associated with a meaningful stimulus through repeated pairing (e.g., Pavlov's dogs associating the sound of a bell with food).
Importance: It helps organisms adapt to their environment by forming connections between stimuli and responses.
Non-Associative Learning:

Definition: Non-associative learning is a change in behavior in response to a single stimulus rather than a pairing of stimuli.
Examples: Habituation, where repeated exposure to a stimulus leads to a decrease in response (e.g., getting used to the sound of a noisy street outside your window).
Importance: It allows organisms to filter out irrelevant stimuli and focus on new or important information.
Pavlovian Conditioning (Classical Conditioning):

Definition: A type of associative learning pioneered by Ivan Pavlov, where a neutral stimulus becomes associated with a meaningful stimulus to elicit a reflexive response.
Example: Pavlov's dogs salivating at the sound of a bell after the bell was repeatedly paired with the presentation of food.
Key Components:
Unconditioned Stimulus (UCS): Stimulus that naturally triggers a response (e.g., food).
Unconditioned Response (UCR): Natural response to the UCS (e.g., salivation).
Conditioned Stimulus (CS): Initially neutral stimulus that, after association with the UCS, triggers a conditioned response (e.g., bell).
Conditioned Response (CR): Learned response to the CS (e.g., salivation in response to the bell).
Hebbian Learning:
Definition: A theory proposed by Donald Hebb stating that when neurons repeatedly fire together, the connection between them strengthens, leading to the formation of neural circuits and memory formation.
Key Principle: "Cells that fire together wire together."
Examples: Learning to ride a bike, where repeated practice strengthens the connections between neurons involved in balancing and coordination.
Importance: It provides a basis for understanding how neural networks in the brain encode information and learn from experiences.
These learning mechanisms play crucial roles in understanding how organisms adapt to their environments, form memories, and acquire new behaviors.

Hypothesis Testing: It's a statistical method used to make inferences about a population parameter based on sample data. The process involves formulating a hypothesis about the population parameter, collecting data, and then using statistical tests to determine the likelihood of observing the results if the null hypothesis were true.

Types of Tests:

T-test: Used to determine if there is a significant difference between the means of two groups.
Chi-square test: Used to determine if there is a significant association between categorical variables.
ANOVA (Analysis of Variance): Used to compare means of three or more groups to determine if they are significantly different from each other.

Significance Level (α): It is the probability of rejecting the null hypothesis when it is actually true. Commonly used levels are 0.05 or 0.01, indicating a 5% or 1% chance, respectively, of making a Type I error.

P-value: It's the probability of obtaining test results at least as extreme as the ones observed during the test, assuming that the null hypothesis is true. A small p-value (typically < 0.05) indicates strong evidence against the null hypothesis, leading to its rejection.

Types of Errors:

Type I Error: Rejecting the null hypothesis when it is actually true (False Positive).
Type II Error: Failing to reject the null hypothesis when it is actually false (False Negative).
Power: It's the probability of correctly rejecting the null hypothesis when it is false. High power indicates a low probability of committing a Type II error.

Effect Size: It measures the strength of the relationship between two variables in a statistical population. It's an important consideration alongside statistical significance.

Normal Probability Curve (or Normal Distribution): It's a bell-shaped curve that represents the probability distribution of a continuous random variable that follows a normal distribution. It's symmetric around the mean, with the majority of values falling close to the mean and fewer values further away. It's characterized by its mean (μ) and standard deviation (σ), and it's commonly used in hypothesis testing and confidence interval estimation due to its mathematical properties.